

PRESS RECORD

COMPX₃₄₁ - Advanced Software Engineering Design



THE UNIVERSITY OF
WAIKATO
Te Whare Wānanga o Waikato

Software Maintenance – Analysis and Testing

COMPX₃₄₁

Overview

- Key issues in software maintenance
- Maintenance process
- Techniques for Maintenance

Key Issues in Software Maintenance

Key Issues in Software Maintenance

- Technical Issues
- Management Issues
- Maintenance Cost Estimation
- Software Maintenance Measurement

Key Issues in Software Maintenance

Technical Issues:

- Limited understanding

“Research indicates that about half of the total maintenance effort is devoted to understanding the software to be modified.”
- Testing

“The cost of repeating full testing on a major piece of software is significant in terms of time and money.”
- Impact Analysis

“Impact analysis describes how to conduct, cost effectively, a complete analysis of the impact of a change in existing software.”

...

Key Issues in Software Maintenance

Technical Issues (continued):

- Impact Analysis

Note, probably not for emergencies

1. analyse the modification request (MR) / problem request (PR);
2. replicate or verify the problem;
3. develop options for implementing the modification;
4. document the MR/PR, the results, and the execution options;
5. obtain approval for the selected modification option.

Key Issues in Software Maintenance

Technical Issues (continued):

- Limited understanding

“Research indicates that about half of the total maintenance effort is devoted to understanding the software to be modified.”
- Testing

“The cost of repeating full testing on a major piece of software is significant in terms of time and money.”
- Impact Analysis

“Impact analysis describes how to conduct, cost effectively, a complete analysis of the impact of a change in existing software.”
- Maintainability

“The capability of the software product to be modified.”

Key Issues in Software Maintenance

Management issues

- Alignment with organizational objectives
 - “the view at the senior management level is often that of a major activity consuming significant resources with no clear quantifiable benefit for the organization.”
- Staffing
 - “maintenance personnel are frequently viewed as “second-class citizens,” and morale therefore suffers”
- Process
 - “Maintenance also requires several activities that are not found in software development ... These activities present challenges to management.”
- Organizational aspects of maintenance
 - “how to identify which organization and/or function will be responsible for the maintenance of software.”
- Outsourcing

Key Issues in Software Maintenance

Maintenance cost estimation

- Cost estimation

“how impact analysis identifies all systems and software products affected by a software change request and develops an estimate of the resources needed to accomplish that change.”

How? By using parametric models and expert experience.

- Parametric Models

“historical data from past maintenance are needed in order to use and calibrate the mathematical models.”

- Experience

“Expert judgment is often used to estimate maintenance effort.”

Key Issues in Software Maintenance

Software Maintenance Measurement

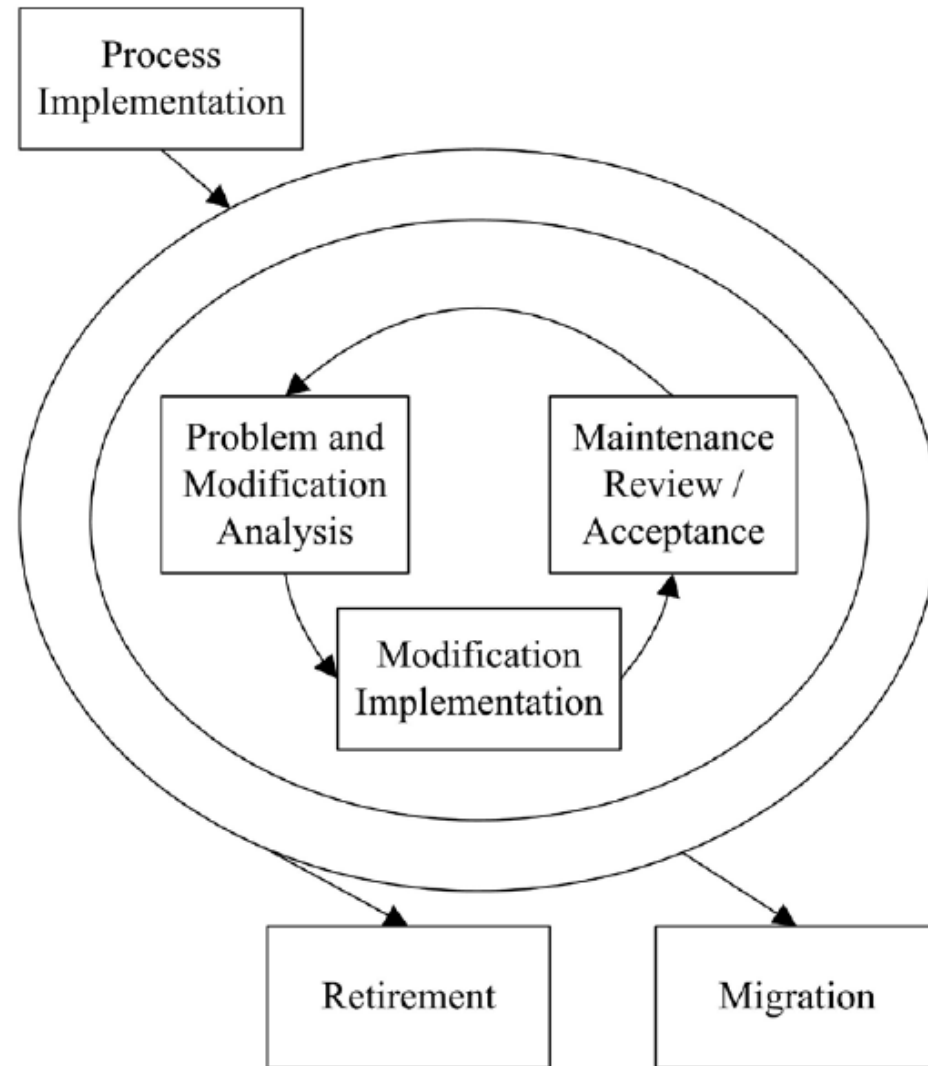
- Analyzability
How difficult (effort and resources) was it for the maintainer to diagnose a problem or identify a modification?
- Changeability
How difficult (effort and resources) was it for a maintainer to implement a modification?
- Stability
How much unexpected behavior did the system exhibit? Before and after modification, including testing.
- Testability
How difficult (effort and resources) was it for the maintainer (and user) to test the modified software?
- Other
Software size, complexity, understandability, maintainability

Maintenance Process

Maintenance process

- The Maintenance lifecycle includes a series of activities, some of which we may not have seen before
 - Process implementation
 - Problem and modification analysis
 - Modification implementation
 - Maintenance review/acceptance
 - Migration
 - Software retirement

Maintenance process



Maintenance process

- Within this lifecycle, some activities will be familiar
 - Analysis
 - Design
 - Coding
 - Testing
 - Documentation
- But some activities are unique to maintenance ...

Maintenance process

Unique Activities

- Program understanding
- Transition
 - Transferring software from developer to maintainer
- Modification request acceptance/rejection
 - Size/effort/complexity of modifications
 - Maintainer versus developer
- Maintenance help desk
 - Assessment, prioritization, and costing of modification requests
- Impact analysis
- Maintenance Service-Level Agreements (SLAs)
 - Contractual obligations between the client and the software provider

Techniques for Maintenance

Techniques for Maintenance

- Program comprehension
- Re-engineering
- Reverse Engineering
- Migration
- Retirement

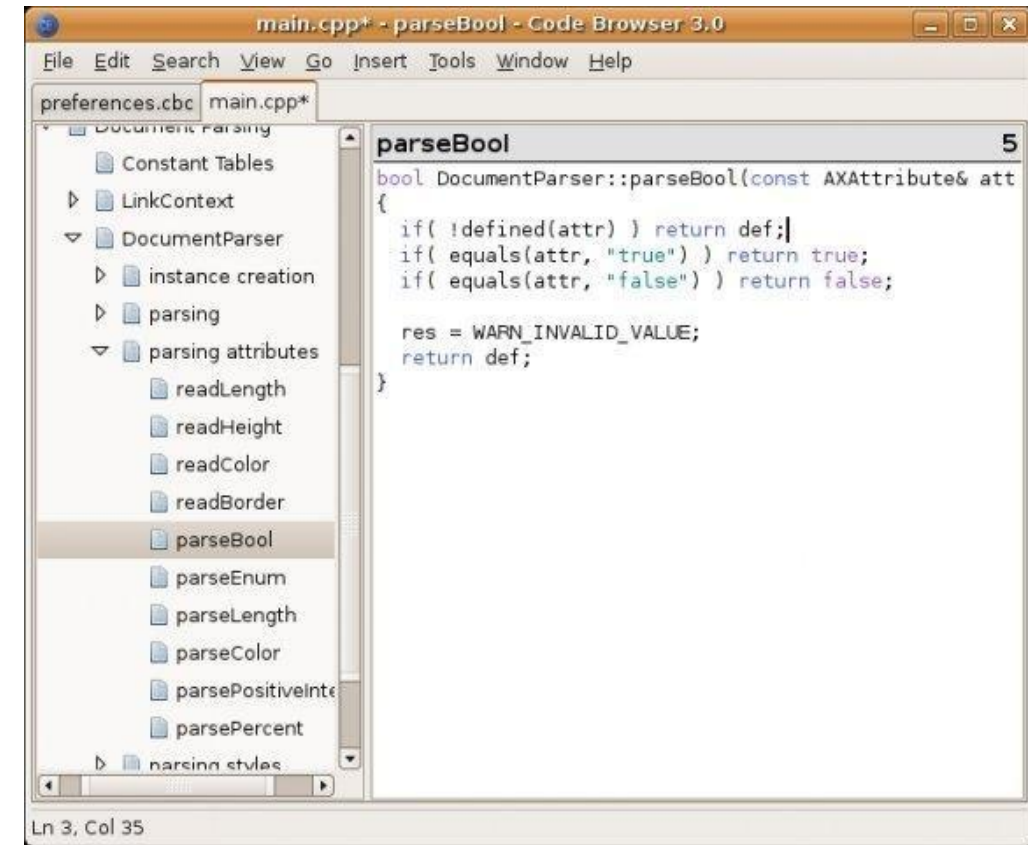
Techniques for Maintenance

Program comprehension

- Technical issue: limited understanding
- Unique activities: program understanding
- Techniques: program comprehension

“Programmers spend considerable time reading and understanding programs in order to implement changes.”

- Code Browser (folding text editor)
- Clear and concise documentation



Techniques for Maintenance

Re-engineering

“Reengineering is defined as the examination and alteration of software to reconstitute it in a new form”

- Used to replace aging or legacy software
- Used to improve program structure and maintainability
- Use refactoring to reorganize a program without changing its behavior

Techniques for Maintenance

Reverse Engineering

“Reverse engineering is the process of analyzing software to identify the software’s components and their inter-relationships and to create representations of the software in another form or at higher levels of abstraction.”

- Produce call graphs and control flow graphs from source code
- Redocumentation
- Design recovery
- Data reverse engineering

Techniques for Maintenance

Migration

“During software’s life, it may have to be modified to run in different environments.”

- Determine the actions needed to accomplish the migration
- Develop and document the steps required
- Migration plan
 - Migration requirements
 - Migration tools
 - Conversion of product and data
 - Execution
 - Verification
 - Support

Techniques for Maintenance

Retirement

“Once software has reached the end of its useful life, it must be retired.”

- Analysis is performed to make the decision to retire
- Retirement plan
 - Analysis
 - Retirement requirements
 - Impact
 - Replacement
 - Schedule
 - Effort
 - Accessibility of archived data

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